

Time & Work — Fundamentals

Work, Rate, Efficiency & the Core Formulas

Aptitude Grind

Why Time & Work Matters for TCS NQT

Time & Work is one of the highest-frequency topics in TCS NQT (Ninja + Digital) Numerical Ability, alongside Percentages and Ratios. Unlike SSC CGL or CAT, TCS rarely asks difficult algebraic work-time puzzles — it sticks to **direct formula-based questions with at most 1–2 twists**: ratio based worker efficiency, pipes & cisterns, alternate working/resting, men-women-children efficiency, and machine problems. The entire topic rests on one idea: *work, rate, and time are locked together by a single equation, and almost every question is just that equation applied from a different angle.*

1 The Golden Rule

Core Identity

$$\text{Work} = \text{Rate} \times \text{Time}, \quad \text{i.e. } W = R \times T$$

- **Work** — the total task, conventionally treated as **1 unit** (a whole job) unless stated otherwise.
- **Rate** — work done per unit time (also called *efficiency* or *one day's/hour's work*).
- **Time** — days/hours taken.

If total work = 1, then $\text{Rate} = 1/\text{Time}$.

This single idea solves almost every question in this topic — everything below is a variation of it.

2 One Day's Work

If **A** finishes a work in N days, A's one day's work is $1/N$.

Example

A completes a work in 8 days $\Rightarrow$ one day's work = $1/8$.

The relationship also runs backward: if A's one day's work is $1/15$, then A alone takes **15 days** to finish the whole job. Time and one-day's-work are reciprocals of each other.

3 Combined Work — Two Workers

If A finishes a job alone in x days and B finishes it alone in y days, their combined one-day's work is $1/x + 1/y$, so together they take:

Combined Work Formula

$$\text{Time (A + B)} = \frac{xy}{x + y}$$

This formula is extremely common and worth memorizing outright.

Worked Example

A = 10 days, B = 15 days.

$$\text{Together} = \frac{10 \times 15}{10 + 15} = \frac{150}{25} = 6 \text{ days}$$

4 Combined Work — Three or More Workers

For three workers:

$$\text{Combined one-day's work} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

Take the LCM of x, y, z to add these cleanly (see `02-fast-tricks-and-shortcuts.tex` for the LCM shortcut). For n workers, the pattern generalizes — just sum every individual rate.

5 The Difference Formula (Finding One Worker From a Pair)

If A alone takes x days, and A + B together take y days, B's one-day's work is:

Difference Formula

$$\frac{1}{B} = \frac{1}{y} - \frac{1}{x}$$

then invert to get B's number of days.

Worked Example

A = 30 days, A + B = 12 days.

$$\frac{1}{B} = \frac{1}{12} - \frac{1}{30} = \frac{5}{60} - \frac{2}{60} = \frac{3}{60} = \frac{1}{20} \Rightarrow B = 20 \text{ days}$$

This “given one worker + the pair, find the other worker” pattern is one of the most frequently tested variants of Time & Work.

6 Efficiency Concept (Very Important)

Inverse Proportionality

$$\text{Efficiency} \propto \frac{1}{\text{Time}}$$

Higher efficiency means less time, and the two are always **inversely proportional**.

If efficiency ratio $A : B = 2 : 3$, then time ratio $A : B = 3 : 2$ — always the reverse of the efficiency ratio.

Example

Time ratio $2 : 5 \Rightarrow$ efficiency ratio $5 : 2$.

6.1 Worker-Ratio Questions — Think in Rates, Not Days

Example: “A is twice as efficient as B.”

Right Approach

Assume work *rates* directly. Let B’s rate = 1 unit/day, then A’s rate = 2 units/day. Never assume days first — assume the rate ratio, since that’s what the problem is actually telling you.

This single habit — assigning rates instead of days whenever a ratio is given — is the fastest problem-solving upgrade in this entire topic.

7 Men, Women, and Children Equivalence

Frequently, a question gives an equivalence like “2 Men = 3 Women” and expects you to convert everyone into one common unit before adding rates.

Worked Example

$2M = 3W$ (i.e. equal work done in equal time)

$$2M = 3W \Rightarrow M : W = 3 : 2 \Rightarrow 1 \text{ man} = \frac{3}{2} \text{ women (in work output)}$$

Always convert every category (men, women, children) into a single common unit of efficiency before combining rates — treat this exactly like the ratio-to-rate trick above.

8 Machine / Printer / Robot / Server Problems

Conceptually **identical** to worker problems — a machine, printer, robot, or server is just another “worker” with its own rate. No new formula is needed; apply everything above directly.

9 Pipes and Cisterns

Same core concept, with a sign convention:

Pipe Sign Convention		
Pipe type	Effect	Rate
Filling (inlet)	Positive work	$+1/x$ (fills in x hours)
Emptying (outlet)	Negative work	$-1/y$ (empties in y hours)

Both pipes open together:

$$\text{Net rate} = \frac{1}{x} - \frac{1}{y}, \quad \text{Time} = \frac{1}{\text{Net Rate}}$$

Worked Example

Fill pipe = 8 hr, Empty pipe = 12 hr.

$$\text{Net rate} = \frac{1}{8} - \frac{1}{12} = \frac{3}{24} - \frac{2}{24} = \frac{1}{24} \Rightarrow \text{Time} = 24 \text{ hours}$$

Common Mistake

Adding the empty-pipe rate instead of subtracting it. Always treat outlet pipes as **negative** contributions to the net rate.

10 Worker-Day Concept

Another foundational identity, used constantly for workers joining/leaving problems:

Worker-Day Identity

$$\text{Total Work} = \text{Workers} \times \text{Days}$$

(only valid when every worker has equal, constant efficiency)

Worked Example

20 workers finish a job in 15 days $\Rightarrow$ Total work = $20 \times 15 = 300$ worker-days.

This “worker-days” total is treated as a fixed, conserved quantity — exactly like total work = 1 in the fractional approach, just scaled up to avoid fractions. See `02-fast-tricks-and-shortcuts.tex` §3 for how this is applied to workers-leaving and workers-joining problems.

11 Wages Are Proportional to Work — Not Time

Very Common Trap

Money $\propto$ Work Done, never time spent. If efficiency ratio is 3 : 5, wages split in the ratio 3 : 5 — **regardless of how many days each person actually worked**, as long as they worked together on the same job. This trips people up constantly because it feels like it should depend on time, but wages track *output*, not *hours logged*.

12 Formula Sheet — Quick Reference

Master Table

Situation	Formula
Work	Rate $\times$ Time
Rate (one day's/hour's work)	Work / Time = 1/Time
Time	1/Rate
Combined work (A + B)	$xy/(x + y)$
Combined work (n workers)	Sum of all individual rates, then invert
Difference formula	$1/B = 1/(A+B \text{ time}) - 1/(A \text{ time})$
Pipe filling	$+1/x$
Pipe emptying	$-1/y$
Efficiency	1/Time (inversely proportional to time)
Wages	$\propto$ Work done, not time
Worker-days	Workers $\times$ Days (constant total, equal efficiency)

13 Key Takeaways

- Everything reduces to **Work = Rate $\times$ Time**, with total work usually normalized to 1.
- Time and rate are always **reciprocals**; efficiency and time are always **inversely proportional**.
- For ratio-based questions (“A is twice as efficient as B”), assign **rates directly** — don’t guess at days.
- Wages split by **work contributed**, never by time spent.
- The worker-day identity (Workers $\times$ Days = constant) is the backbone of every joining/leaving problem — see `02-fast-tricks-and-shortcuts.tex`.